Identifying Undervalued U.S. Housing Markets for Real Estate Investment

Problem Statement

Real estate investors and investment firms must decide where to allocate capital to maximize return while managing risk. However, housing markets vary widely in price growth, rental demand, affordability, population trends, and economic fundamentals. The goal of this project is to build a data-driven model that identifies U.S. housing markets that are potentially undervalued today but show strong indicators of future growth. The analysis will provide actionable recommendations on where an investor could consider purchasing property to achieve above-average appreciation or rental performance.

Context

The U.S. housing market has experienced significant volatility due to interest rate changes, migration patterns, remote work, and supply constraints. Investors increasingly rely on data to identify emerging markets before prices fully adjust. A model that combines historical home prices, economic indicators, and regional demand signals can help investors make more informed and objective decisions.

This project simulates the role of a data scientist working for a real estate investment firm that wants to prioritize markets for expansion.

Criteria for Success

The project will be successful if it:

- Identifies key features that drive home price growth
- Builds a model that can predict future price appreciation or market strength
- Produces a ranked list of attractive housing markets
- Provides clear, business-focused recommendations supported by data

Model performance will be evaluated using appropriate regression metrics such as MAE and R^2.

Scope of the Solution Space

This project will:

- Focus on U.S. metropolitan or county-level housing markets
- Use historical data to predict future growth potential
- Compare markets based on investment attractiveness

It will not:

- Attempt to forecast short-term home prices at the individual property level
- Include property-level operational costs or financing strategies

The goal is **market selection**, not property valuation.

Constraints

Potential constraints include:

- Data availability and consistency across regions
- Time limitations for feature engineering and model tuning
- Housing data that may lag current market conditions

These will be addressed through careful dataset selection and clear documentation of assumptions.

Stakeholders

The primary stakeholder is a **real estate investment firm** seeking data-driven guidance on where to deploy capital.

Secondary stakeholders include:

- Portfolio managers
- Market research teams
- Individual real estate investors

They care because the output will directly influence investment strategy and potential returns.

Data Sources

Potential data sources include:

- Zillow Research housing price data (home value index, rent index)
- U.S. Census economic and population data
- Federal Reserve economic indicators
- Kaggle datasets containing regional housing and demographic data

These datasets provide historical price trends and the economic drivers behind housing demand.

Proposed Approach

The project will follow the full data science methodology:

1. Data wrangling and cleaning
2. Exploratory data analysis to identify trends and relationships
3. Feature engineering to capture growth indicators (price momentum, affordability, migration, etc.)
4. Model development using regression techniques

5. Model evaluation and interpretation
6. Translating results into investment recommendations

Deliverables

The final outputs will include:

- A GitHub repository containing all code and documentation
- A written project report
- An executive-style slide presentation with business recommendations